Aditi Bhole

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.30
Atomic Energy Central School Kakrapar	Class XII (CBSE)	2019	83.80%
Atomic Energy Central School Kakrapar	Class X (CBSE)	2017	9.80%

WORK EXPERIENCE

RELIANCE INDUSTRIES LIMITED | Graduate Engineer Trainee - Process Design | Navi Mumbai

- Supported process design and engineering activities, including hydraulic calculations and pressure-drop analysis using KORF.
- Reviewed P&IDs, process documents and engineering deliverables to understand process systems, equipment and piping configurations.
- Assisted with technical document review and process engineering activities, supporting the development and verification of process design deliverables

PROJECTS

Adaptive Satellite-Based Forest Fire Detection | *Python, PyTorch, CNN,ResNet, ViT, Swin Transformer, Agentic AI*

- Built a deep learning pipeline for satellite-based wildfire classification, implementing a custom CNN with automated preprocessing, data augmentation, GPU-accelerated training, and validation-driven model selection.
- Developed an Agentic AI inference router that adaptively selects Tiny/Small/Large vision models based on input complexity and computational constraints, optimizing accuracy, latency, and resource utilization.
- Established a multi-objective neural architecture search and benchmarking framework across CNN and Transformer architectures, jointly optimizing accuracy, model complexity, and inference latency for scalable real-time wildfire detection.

AI-Assisted Systematic Literature Review | *Python, LLMs, Pandas, UMAP, HDBSCAN*

- Engineered an LLM-driven literature mining pipeline for large-scale bibliographic screening, performing semantic relevance assessment, metadata extraction, and methodological classification from Title-Abstract-Keywords.
- Implemented high-dimensional LLM embedding generation, UMAP dimensionality reduction, and HDBSCAN density-based clustering to identify latent research themes without predefined cluster counts.
- Developed an automated LLM-based semantic taxonomy and classification framework for mapping publications into methodological and application-specific research domains, enabling trend analysis and emerging-topic discovery.

Graph Neural Network-Based Edge Importance Learning for Combinatorial Route Optimization | *Python, NumPy, GCN, Graph Learning, 2-opt*

- Constructed a 100-node spatial graph with 9,900 directed edges using KNN connectivity and Euclidean distances, followed by a 2-layer GCN for graph-aware node representation learning.
- Generated 30 pseudo-optimal routes using Nearest Neighbor and 2-opt, producing 297K route-edge training samples with binary edge labels for supervised learning.
- Designed a 33D edge feature space from source/destination node embeddings and distance, and trained a logistic edge-importance model to generate learned edge scores and a cost matrix for downstream route optimization.

Gaussian-ANN: Machine Learning Adaptation of Gaussian Wake Model | *Python, Artificial Neural Networks, Wind Energy, Wake Modelling*

- Integrated a feed-forward ANN with the Jensen analytical wake model to learn nonlinear wake-radius expansion as a function of downstream turbine distance.
- Formulated a hybrid ANN-Jensen framework where the ANN-predicted wake radius is propagated through the Jensen model to estimate velocity deficits and downstream wind velocity.
- Implemented customized backpropagation gradients by applying the chain rule through the wake-radius-to-velocity mapping, enabling ANN parameters to be trained on a velocity-deficit loss despite wake radius being the network output.

SKILLS

Programming: Python, SQL, C, MATLAB

AI/ML: Machine Learning, Deep Learning, Computer Vision, Transformers, GNNs, Agentic AI, Optimization

Data & Tools: NumPy, Pandas, SciPy, Scikit-learn, Microsoft Office, Git/GitHub, Jupyter, Power BI

Soft Skills: Communication, Analytical Thinking, Teamwork

POSITIONS OF RESPONSIBILITY

Placement Coordinator | Office of Career Services, IIT Hyderabad -- Served as a Company Coordinator, proactively pitching IIT Hyderabad to prospective recruiters and managing recruiter-student communication, and placement activities.

Technical Head | Chemical Engineering Students Association, Nirma University -- Led technical initiatives, workshops and student activities while coordinating with faculty and student teams.

PUBLICATIONS

- Biowaste derived UV-Visible-NIR active Z-scheme CaO/MoS photocatalyst as a low-cost, waste-to-resource strategy for rapid wastewater treatment -- Journal of Photochemistry & Photobiology A: Chemistry, 2024.
- Treatment of waste using waste-derived materials and free energy: A practical concept of circular-economy -- Sustainable Chemistry and Pharmacy, 2024.

SCHOLASTIC ACHIEVEMENTS

Amazon ML Summer School 2026| Amazon - Selected among 3,000 participants from 1,34,421 applicants for the Amazon ML Summer School 2026, a selective program focused on Machine Learning, Deep Learning, and advanced AI concepts.